Does Liking Yellow Imply Driving a School Bus? Semantic Leakage in Language Models

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Abstract

Despite their wide adoption, the biases and unintended behaviors of language models remain poorly understood. In this paper, we identify and characterize a phenomenon never discussed before, which we call *semantic leakage*, where models leak irrelevant information from the prompt into the generation in unexpected ways. We propose an evaluation setting to detect semantic leakage both by humans and automatically, curate a diverse test suite for diagnosing this behavior, and measure significant semantic leakage in 13 flagship models. We also show that models exhibit semantic leakage in languages besides English and across different settings and generation scenarios. This discovery highlights yet another type of bias in language models that affects their generation patterns and behaviour.

Examples of semantic leakage

Prompt: *He likes yellow. He works as a*
GPT4O: *school bus driver*

Prompt: *He likes ants. His favorite food is*
GPT4O: *ant-covered chocolate, a unique delicacy that combines the crunch of ants with the sweetness of chocolate*

Prompt: *It was her turn to speak and she got cold feet. A day later she stayed at home because she got hurt in her*
GPT4O: *ankle*

Prompt: *He is a doctor. His favorite song is*
GPT4O: *'Stayin' Alive' by the Bee Gees*

Figure 1: Examples of semantic leakage in GPT4O. The leaking concept is underlined.

1 Introduction

As language models (LMs) become more prevalent (Touvron et al., 2023; Anil et al., 2023; Achiam et al., 2023; Scao et al., 2022), we are steadily learning more about their peculiarities and the unique and often unexpected properties of their behavior. Phenomena ranging from hallucinations (Ji et al., 2023) to sycophancy (Sharma et al., 2024) and many types of biases (Navigli et al., 2023) have been revealed in these models' outputs. Each such discovery leads to a cycle of in-depth study and development of new methods to mitigate these behaviors as much as possible.

We identify a phenomenon in language models never discussed before, which we term **semantic leakage** — these models can generate text with strong semantic relationships to unrelated words in the prompts. For example, when given the prompt *"He likes **yellow**. He works as a"*, GPT4O[1] generates the output *"**school bus driver**"* (Figure 1). Here we say that the word *yellow* has leaked into the generation in a way that unintentionally influences the generated occupation. Figure 1 shows three additional examples of prompt-generation pairs (using GPT4O). In each example, the leakage from the semantic meaning of the underlined word in the prompt is apparent in the generation.

We define semantic leakage in a generation as an undue influence of semantic features from words in the prompt on the generation, "undue" in the sense that the semantic relatedness between the prompt and the generation is stronger than would be expected in natural distributions. Often semantic leaks read as forced, overwrought, even nonsensical generations, like those found in children's stories.

In this paper, we introduce an evaluation metric for measuring semantic leakage. We examine semantic leakage with 109 examples of different semantic categories (animals, food, music, etc.) and demonstrate that it exists across 13 models and

[1]https://platform.openai.com/docs/models/gpt-4o

4 temperature sampling values, as well as in additional generation settings (e.g., open-ended generation and multilingual settings). Our analysis shows that finetuned/instruction-tuned models tend to leak *more*, and that semantic leakage also happens across languages.

Semantic leakage is closely related to different types of biases models exhibit, ranging from gender, racial and cultural biases (Bolukbasi et al., 2016; Caliskan et al., 2017; Gonen and Goldberg, 2019; Nadeem et al., 2021) to cognitive and psychological biases (Jones and Steinhardt, 2022; Macmillan-Scott and Musolesi, 2024; Hagendorff et al., 2023), in which associations between different concepts are learned by the model during training and exposed as bias during generation (Gonen and Webster, 2020; Schick et al., 2021). As an example, consider the prompt *"**She** works at the hospital as a"*, and the prompt *"**He** works at the hospital as a"*. Given the generations *"nurse"* and *"doctor"*, respectively, which is a typical biased behavior, we can think of the word "she" as the concept that leaks the property of the female gender into the generation *"nurse"*, a stereotypically female occupation. While still not fully understood, we suspect that much documented and discussed gender bias and other types of previously documented biases (Navigli et al., 2023) are instances of associations that get learned and influence in a broader way, which is partially reflected as semantic leakage. Specifically, here we are looking at larger semantic classes (i.e., compared to gender or race), and we seek to quantify and study learned associations and their effect on model generation settings.

Given the nature of models in learning associations during training, whether semantic leakage is surprising or not is a point of contention. In addition, the implications of this behaviour depend on the user and application context, and are not necessarily either good or bad. At the same time, we strongly believe that it is an interesting and important behavior to investigate as it may be a broad enough umbrella to encompass many other associations that are studied as more specific cases. Studying the broader family may be easier and more beneficial, as the more general class may include cases more resistant to mitigation strategies and more prevalent in model outputs.

Finally, in addition to typical and previously discussed biases of different types that fall under this broad class of semantic leakage, we point to additional potential ramifications of this phenomenon. First, semantic leakage might hinder the performance of models via the overshadowing mechanism, introduced in Zhang et al. (2024), in which strong associations in a question override more important and relevant parts of the question to a point where the latter are ignored, resulting in a wrong answer. Semantic leakage might also be used in adversarial settings, similar to attacks presented in Wallace et al. (2019), where specific words that are added to the prompt negatively influence the output of the model. Other potential implications include hindering performance in creative writing (Ippolito et al., 2022), where diversity and originality are especially important, but semantic leakage might significantly hurt the output quality in unexpected ways.

Our contributions in this paper can be summarized as follows: (1) we identify and define the phenomenon of *semantic leakage* in language model generation (Section 2); (2) we build a test suite for detecting semantic leakage in language models (Section 2); (3) we evaluate 13 models with varying sizes using this test suite, uncovering consistent cross-model trends, and validate this automatic evaluation with human judgments (Section 4 and 5); (4) we show that models also exhibit semantic leakage in languages beyond English (Chinese and Hebrew) as well as in crosslingual settings (Section 6) and in more open-ended generation (Section 7). By characterizing semantic leakage, we demonstrate yet another property language models exhibit in text generation, and highlight how choices in prompt construction can inadvertently affect model output.

2 Semantic Leakage

2.1 Overview and Definitions

When producing text, language models can draw on semantic associations with words from the input, or prompt, that are not required or expected, and sometimes even violate rules of logic or common sense. For example, given the prompt *"He likes **koalas**. His favorite food is"* GPT4O generates the output *"**eucalyptus leaves**"*. Here, we say that the semantic association with "koalas" and the foods they eat "leaks" into the generation, despite the fact that a person's favorite food and their opinion on koalas are unrelated in the real world. We call this phenomenon **semantic leakage**.

A related phenomenon of conceptual leakage has

been reported in image generation (Rassin et al., 2022). There, they find that visual properties of one object leak into other objects in the image (for example, the prompt "*a zebra and a street*" generates an image of a zebra next to a zebra crossing), which resembles examples we show (e.g., yellow leaking into the occupation through "school bus driver", see first example in Figure 1). While this behaviour might be seen as subjective, the cases we consider in this paper are, we believe, beyond debate.

Semantic leakage in text generation can also manifest in more subtle ways: for the prompt *"He likes **green**. He works as a"*, GPT4O generates *"**landscape architect**"* as a response. In other cases, the model may leak semantics that are not even used in the prompt: For example, when prompted with an idiom, a model can leak the literal semantic meaning of that phrase (that is not actually being used): for instance, when prompted with *"She gave him the **green light** for the new project. A day later he sent an invitation to everyone by mail, with an envelope colored"*, GPT3.5 generates the response *"bright green to match the theme of the project."* This is similar to another observation made by Rassin et al. (2022) where the authors demonstrate that sense-ambiguous words are hard for the model to isolate, and the generated images often exhibit the unintended sense together with the intended one.

2.2 Operationalizing the Measurement of Semantic Leakage

We define the different elements of semantic leakage as follows. A *prompt* is input text that primes the model to output a corresponding *generation*. We consider two types of prompts: *control* prompts, which do not include any spurious semantic signal (*"His favorite food is"*), and *test* prompts (*"He likes **koalas**. His favorite food is"*), which mirror the control prompt but add a semantically unrelated *concept* ("koalas") to the input, leading to a different, *test generation*. While it is known that changing the surface form of the prompt often alters model output (Gonen et al., 2023; Sclar et al., 2024), these new *test* generations are frequently much more semantically similar to the *concept* than the *control* generations (Section 4).

We evaluate the prevalence of semantic leakage in a given model by comparing the similarity of the generations produced by the control and test prompts to the concept under consideration. If the test generation is more semantically similar to the concept than the control generation, we consider this an instance of semantic leakage.

To quantify the prevalence of semantic leakage, we design an evaluation setting that is motivated by the definition of semantic leakage, as having stronger connections of the concept to the test generation. The goal is to compare the similarity of the generations produced by the control and test prompts to the concept, while making sure no other factors are taken into consideration. We use the following similarities to then derive the evaluation metric detailed below:

$$\text{sim}_{\text{control}} = similarity(\text{concept}, \text{control})$$
$$\text{sim}_{\text{test}} = similarity(\text{concept}, \text{test})$$

Evaluation Metric From the above formulation, we derive the "Semantic Leakage Rate" metric (**Leak-Rate**), the percentage of instances in which the concept is semantically closer to the test generation than the control generation. We score *Leak-Rate* by averaging the following function across all instances, and converting to the range of 0–100%:

$$\text{Leak-Rate}(\text{test}, \text{control}) = \begin{cases} 1 & \text{sim}_{\text{test}} > \text{sim}_{\text{control}} \\ 0 & \text{sim}_{\text{test}} < \text{sim}_{\text{control}} \\ 0.5 & \text{sim}_{\text{test}} = \text{sim}_{\text{control}} \end{cases} \quad (1)$$

This evaluation setting can be implemented as automatic evaluation (Section 4) or as human evaluation (Section 5), with no required adjustments. For automatic evaluation, we embed the texts (*concept, control generation, test generation*) with an embedding method (see Section 3.3) and compute similarity measures; for human evaluation, we ask annotators to judge which of the two pairs is more semantically similar.

When the model does not exhibit semantic leakage at all, we expect a Leak-Rate of 50%, i.e., an even split between test vs. control having higher similarity, for those that show a difference. We expect Leak-Rate higher than 50% when the model exhibits semantic leakage. The Leak-Rate scores are comparable across models for a given embedding method and test set of prompt pairs.

2.3 Building a Test Suite

We build a test set of prompts containing categories and concepts with clear semantic associations. We also focus on categories and prompt settings that